

AI StudyBuddy - AI assistant for smarter learning and study support.

Introduction Project Title: **AI STUDY BUDDY API**

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Abstract:

AI Study Buddy is a full-stack, API-driven learning companion application that helps students study more effectively using artificial intelligence. Users can register, log in, upload their own study materials, and instantly generate AI-powered flashcards, quizzes, and personalized study plans based on the uploaded content. The system reduces manual effort in creating revision material and helps learners prepare for exams in a structured, time-bound manner.

Introduction:

Students often spend a large amount of time manually converting textbooks, notes, and articles into revision material such as flashcards and quizzes. AI Study Buddy automates this process by allowing a user to upload study material once, after which the system uses AI to

generate flashcards, multiple-choice quizzes, and a day-wise study plan tailored to the user's available time and goals. The application exposes all functionality through a REST API, tested using Postman, with a MongoDB database for persistent storage of users and materials.

Objectives:

1. To provide secure user registration and login using token-based authentication.
2. To allow users to upload study materials (notes, text files) to the system.
3. To automatically generate AI-powered flashcards from uploaded material.
4. To automatically generate AI-powered quizzes with multiple-choice questions and answers.
5. To generate a personalized, goal- and time-based AI study plan.
6. To store user and material data reliably in a database for future retrieval.

Scope of the Project:

The system is designed for students and self-learners who want an AI-assisted way to revise any subject. It can be used to convert lecture notes, textbook chapters, or reference documents into ready-to-use study aids, and to plan revision schedules ahead of exams. The scope covers user account management, material upload, and three AI-generation features: flashcards, quizzes, and study plans.

Modules:

- User Authentication Module – Handles user registration and login with access and refresh tokens.
- Study Material Upload Module – Allows users to upload study files (e.g., notes, text documents) linked to their account.
- AI Flashcard Generation Module – Generates questions–answers flashcards from uploaded material using AI.
- AI Quiz Generation Module – Generates multiple-choice quiz questions with options and correct answers.

- AI Study Plan Generation Module – Generates a day-wise study plan based on a stated goal, hours per day, and number of days.
- Database Module – Stores users, study materials, and related data in MongoDB.

System Requirements:

Hardware Requirements:

- Computer or Laptop
- Minimum 4 GB RAM
- Internet connection

Software Requirements:

- Operating System: Windows 10 or above
- Node.js runtime and Express.js framework
- MongoDB Database (viewed via MongoDB Compass)
- Postman (for API testing)
- Web Browser (Chrome/Edge)

System Architecture:

The system follows a client-server architecture exposed entirely through a REST API. The client (tested via Postman in this phase) sends HTTP requests to a Node.js and Express.js backend server. The backend authenticates users using JWT-based access and refresh tokens, processes uploaded study material, calls AI generation logic to build flashcards, quizzes, and study plans, and persists all user and material data in a MongoDB database.

Database Design:

Database: Ai_studybuddy (MongoDB)

Collection: users

- `_id` (Primary Key)
- `name`
- `email` (unique index)
- `password` (hashed)
- `role`
- `createdAt` / `updatedAt`

Collection: materials

- `_id` (Primary Key)
- `user` (reference to users collection)
- `title`
- `content`

Implementation:

The backend is implemented using Node.js with the Express.js framework, and MongoDB is used as the database. Authentication is implemented using access and refresh tokens returned on registration and login. Uploaded study material is stored in the materials collection and linked to the uploading user. AI generation endpoints take the uploaded material's ID as input and return structured flashcards, quiz questions, or a study plan as JSON.

MongoDB Output – Users Collection Indexes:

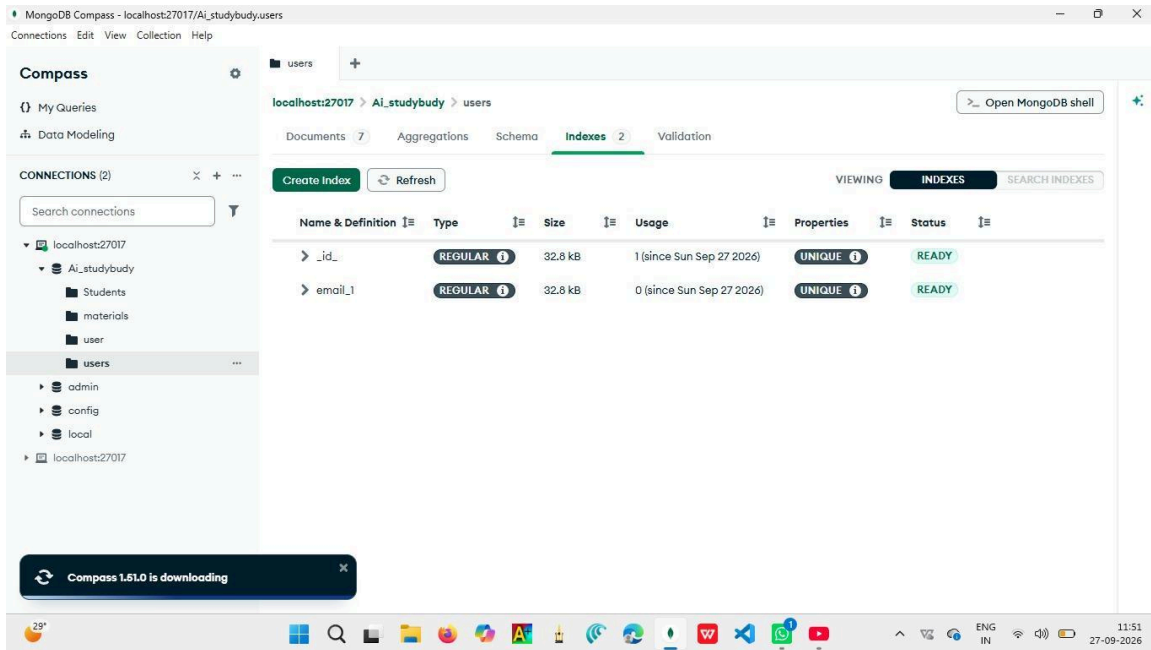


Figure: MongoDB Compass – indexes on the users collection

MongoDB Output – Users Collection Documents:

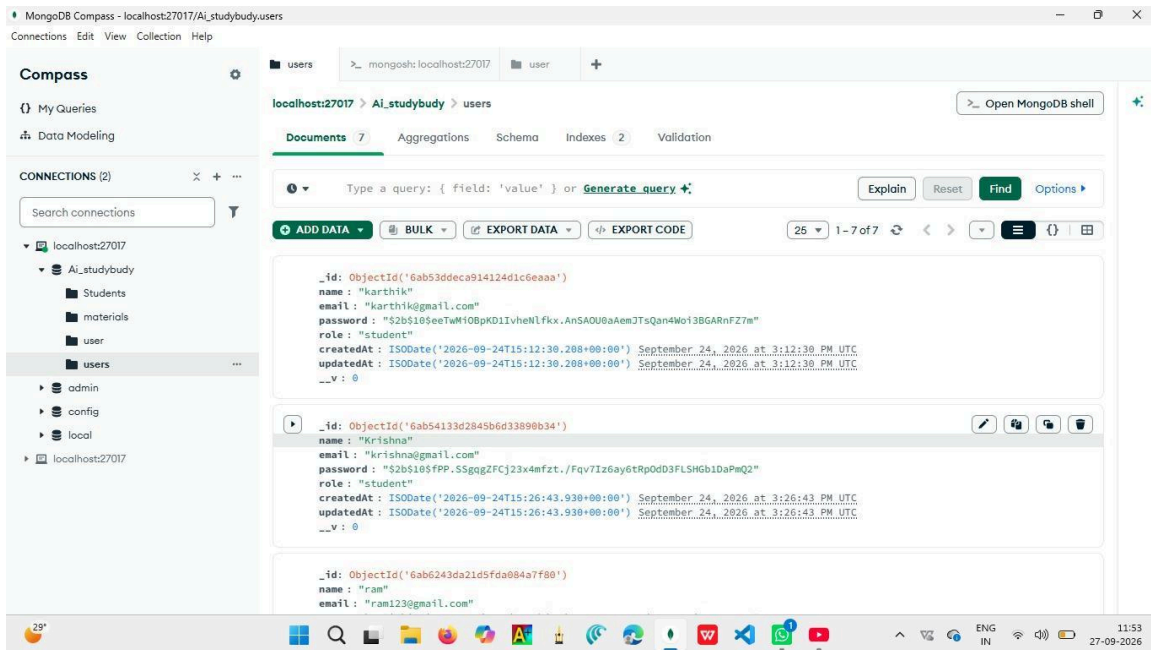


Figure: MongoDB Compass – sample user documents

Testing:

Testing was performed using Postman to verify that all API modules function correctly, including user registration, login, material upload, and the three AI generation endpoints (flashcards, quiz, and study plan).

User Registration Output:

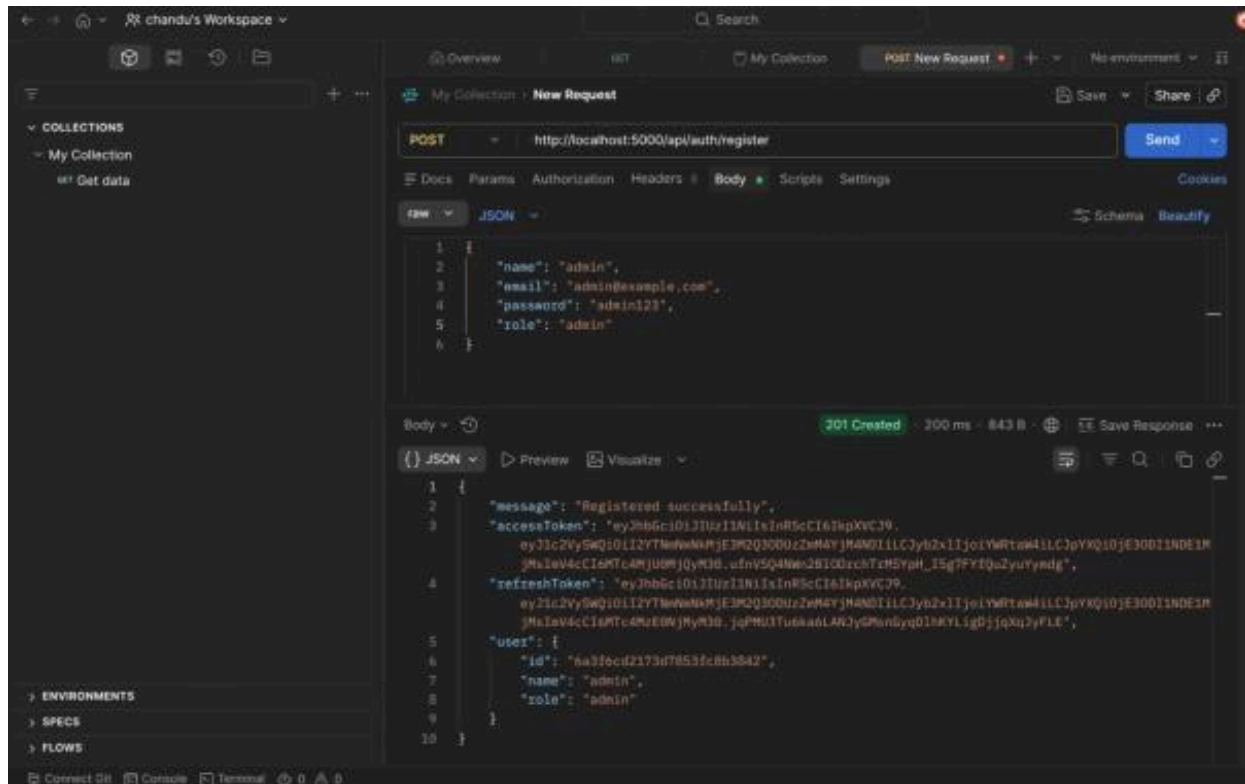


Figure: POST /api/auth/register – 201 Created

User Login Output:

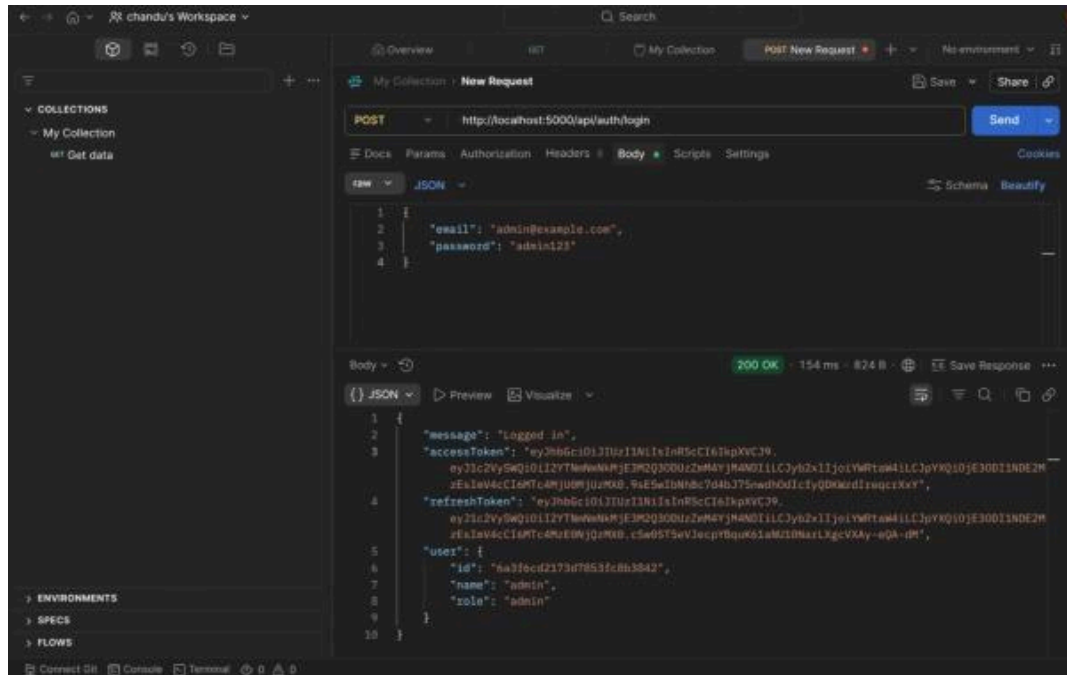


Figure: `POST /api/auth/login` – 200 OK

Study Material Upload Output:

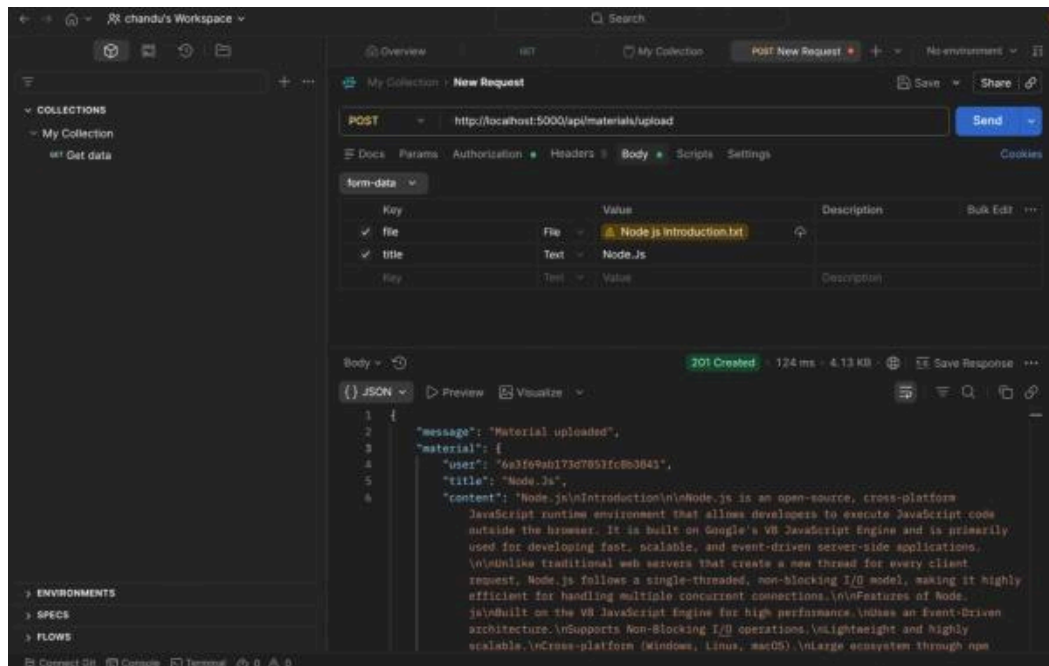


Figure: `POST /api/materials/upload - 201 Created`

Generate AI Flashcards Output:

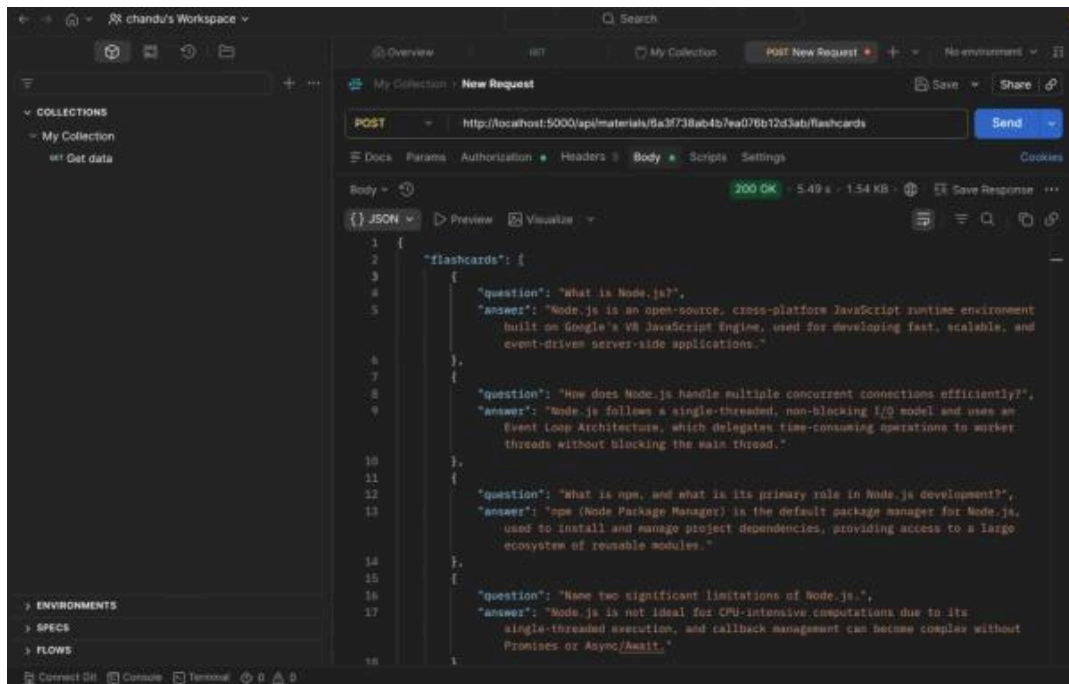


Figure: `POST /api/materials/{id}/flashcards` – 200 OK

Generate AI Quiz Output:

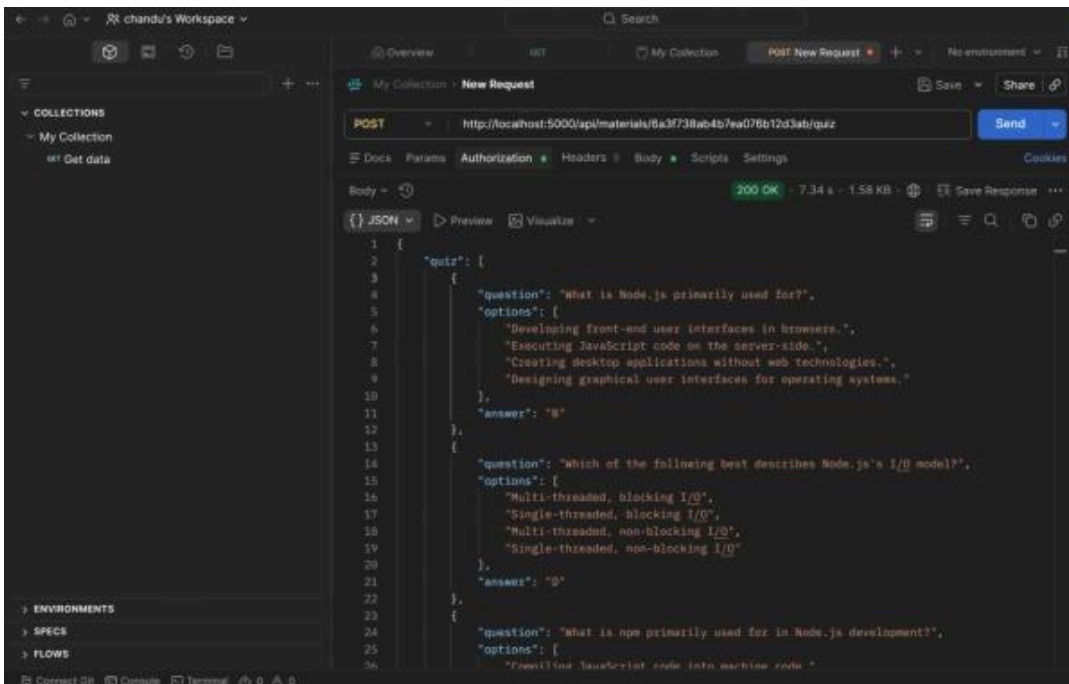


Figure: `POST /api/materials/{id}/quiz` – 200 OK

Conclusion:

AI Study Buddy successfully demonstrates a working, API-driven study assistant that combines secure authentication, material upload, and AI-based generation of flashcards, quizzes, and study plans. Testing through Postman confirms that all core endpoints function as expected, and MongoDB Compass verification confirms that user and material data are correctly persisted. The project provides a solid foundation for building a full front-end study companion application.

Future Enhancements:

- Add a mobile/web front-end interface for end users.
- Support uploading PDF and image-based study material with OCR.
- Track quiz scores and flashcard review progress over time.
- Add reminders/notifications for the generated study plan.

Screen recorder link:

 **NM.mp4**